Zero-shot motion categorization: where we landed

Approach

Each scene is classified purely by cosine similarity between its video embedding and a set of candidate text captions (InternVideo2-Stage2-1B zero-shot, `classify_video()` in `videotag/pipeline.py`). No pixel statistics or other signals - caption matching only.

Final scheme

3 motion tiers - **static / moderate / high** - each backed by 2-3 generic, motion-focused caption phrasings (`videotag/bitrate_categories.py`). The winning tier maps to a bitrate range via a per-resolution table.

On our 101-scene test corpus this gives an **11 / 63 / 27** split (static/moderate/high), which tracks the real spread of motion in the content. This was the best distribution found after testing single-caption tiers, finer-grained tiers, and content-specific captions - all of which collapsed the classification (e.g. zero scenes ever hit "high," or one caption dominating regardless of actual motion). 3 tiers with a few generic phrasings each is the practical ceiling for what this model reliably discriminates via caption matching.

Promise against a real scene/bitrate database

This is a single cheap forward pass per scene (no decode-and-measure, no encoder trial runs) that recovers a 3-way motion split correlated with how much a per-title encoder would spend on a scene. To turn it into a real pre-classification signal:

1. **Calibrate the bitrate table** using real (scene, measured bitrate) pairs instead of the current generic streaming heuristic - a lookup-table change only, doesn't touch the classifier.
2. **Validate on a larger, more varied corpus** than one trailer, to confirm the 3-tier split holds across content types and to get real per-tier bitrate medians for point 1.

With those two steps, this becomes a viable coarse first-pass bitrate-ladder selector ahead of full per-title encoding.